

#PREDICTIVE ANALYTICS

Data mining -

- Process of discovering patterns, correlations, trends, and useful information from large sets of data using statistical, mathematical and computational techniques

- Application:

(i) Retail

(ii) Finance

(iii) Healthcare

(iv) Communication

(v) Manufacturing

source
dataset

data prep.
modeling

output & eval.

prim (predict
analy /
rule
inductio)

Graph Pad

prim

= st. analy.
& visual.

Preprocessing steps

KDD - Knowledge Discovery Database -

Knowledge Discovery (KDD) Process:

- 1) Data cleaning: To remove noise and inconsistent data
- 2) Data integration: To combine multiple source
- 3) Data selection: To retrieve data for analysis
- 4) Data transformation: To transform data into appropriate form for data mining

5) Data mining

6) Evaluation

7) Knowledge presentation

col (Gender) == "M"

If (col (Gender) == "M", "Male",
"Female")

Data Mining Techniques -

- 1) Classification } supervised
- 2) Regression } supervised
- 3) Clustering → unsupervised

excel
format
.xlsx

Phases of Data Mining Process -

- 1) Data Collection
- 2) Data Preprocessing
 - Removing noise and inconsistencies
 - Handling missing values & outliers
 - Transforming & normalizing data to make it suitable for analysis
- 3) Data Integration
- 4) Data Selection
- 5) Data Transformation
- 6) Data Mining
- 7) Pattern Evaluation
- 8) Knowledge Presentation

KDD vs Data Mining -

- KDD is the overall process of extracting knowledge from data while Data Mining is a step inside the KDD process, which deals with identifying patterns in data.
- Data Mining is only the application of a specific algorithm based on the overall goal of KDD process

Binning methods for Data Smoothing -

- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 30
- Partition into (equi depth) bins:

Bin 1: 4, 8, 9, 15
 Bin 2: 21, 21, 24, 25
 Bin 3: 26, 28, 29, 30

} Equal sized bins (4)

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- Smoothing by bin means:

Bin 1: 9, 9, 9, 9 Avg = 9
 Bin 2: 23, 23, 23, 23 Avg = 22.75
 Bin 3: 29, 29, 29, 29 Avg = 29.25

} Mean of the 3 bins

Replace each value in the bin with the closest boundary value (either min/max of that bin)

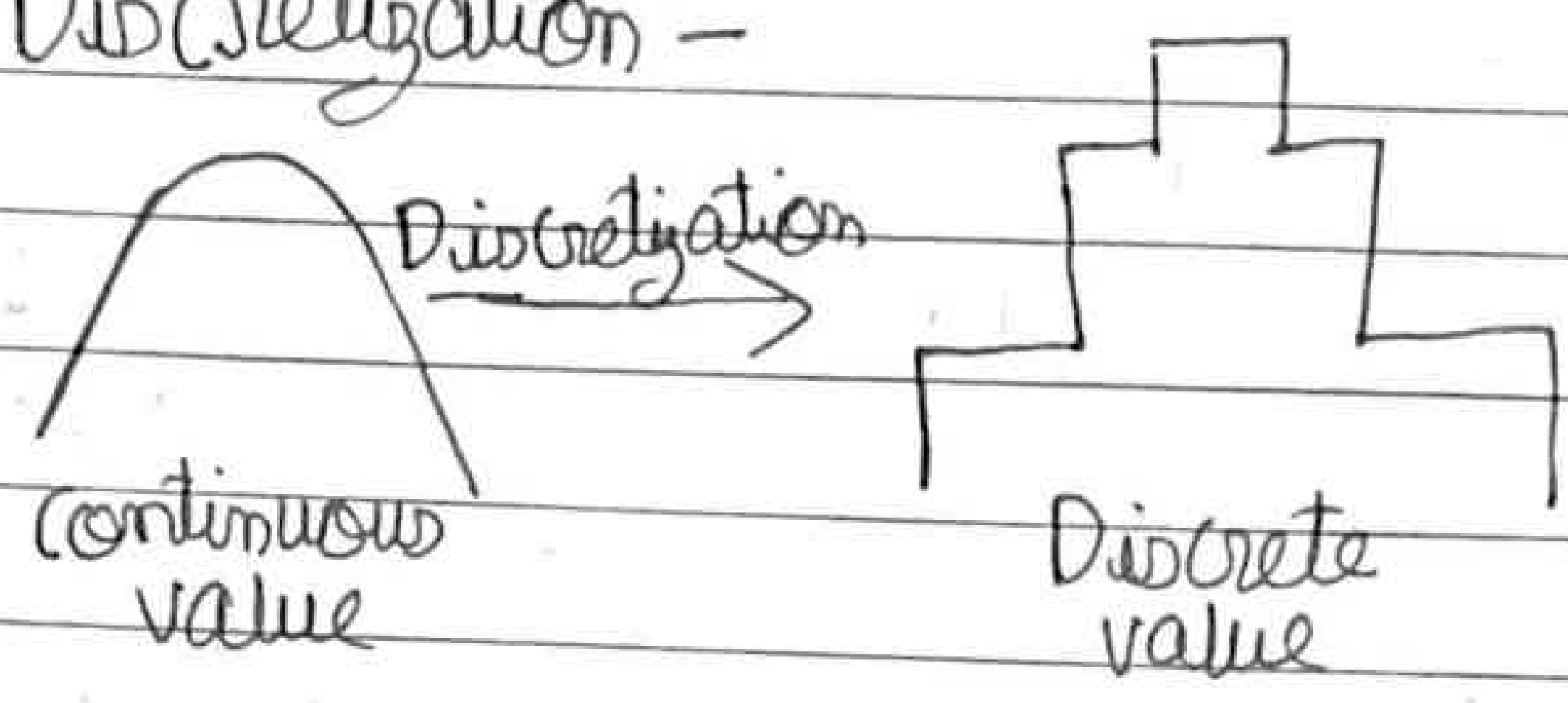
there are near boundary no.

4 4 4 15
↑ ↑ ↑ ↑
4, 8, 9, 15

Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15 Min=4, Max=15
- Bin 2: 21, 21, 25, 25 Min=21, Max=25
- Bin 3: 26, 26, 26, 34 Min=26, Max=34

Discretization -



3 types of attributes:

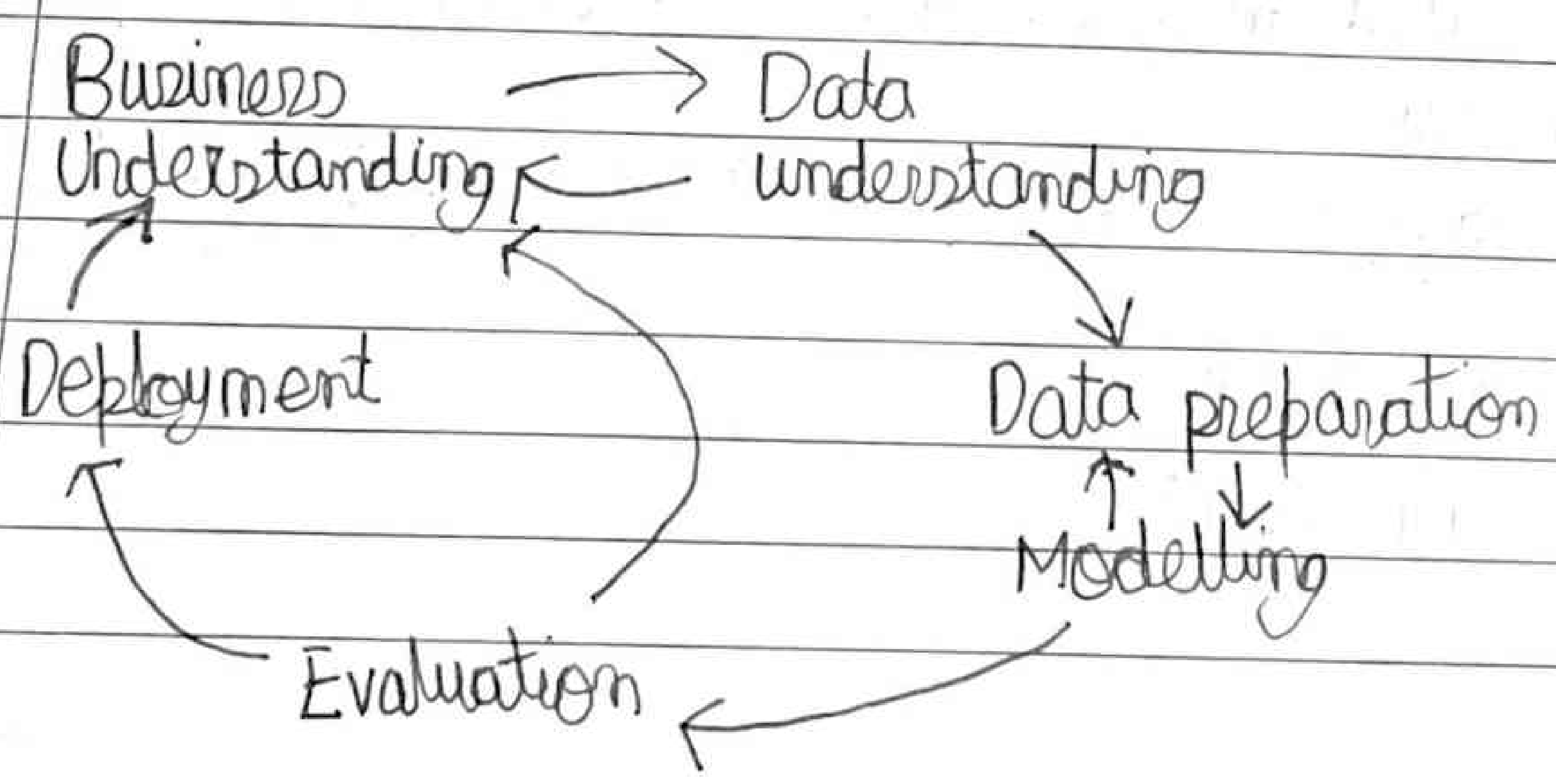
- 1) Nominal = values from an unordered set
- 2) Ordinal = values from an ordered set
- 3) Continuous = real numbers

* * # CRISP-DM (Cross Industry Standard Process for Data Mining)

Widely used data mining process model that provides a structured approach to planning and executing a data mining project

Phases:

- 1) Business Understanding (Understand project goals)
- 2) Data understanding (Collect & explore data)
- 3) Data preparation (Prepare final dataset for modeling)
- 4) Modeling (Apply various modeling techniques)
- 5) Evaluation (Assess the model)
- 6) Deployment (Deliver the data mining results)



Evaluation - Actual

3
10

		(1) Positive	(0) Negative	
Predicted	Positive (1)	TP	FP	→ TN Type I Error
	Negative (0)	FN	TN	

↓
Type II Error

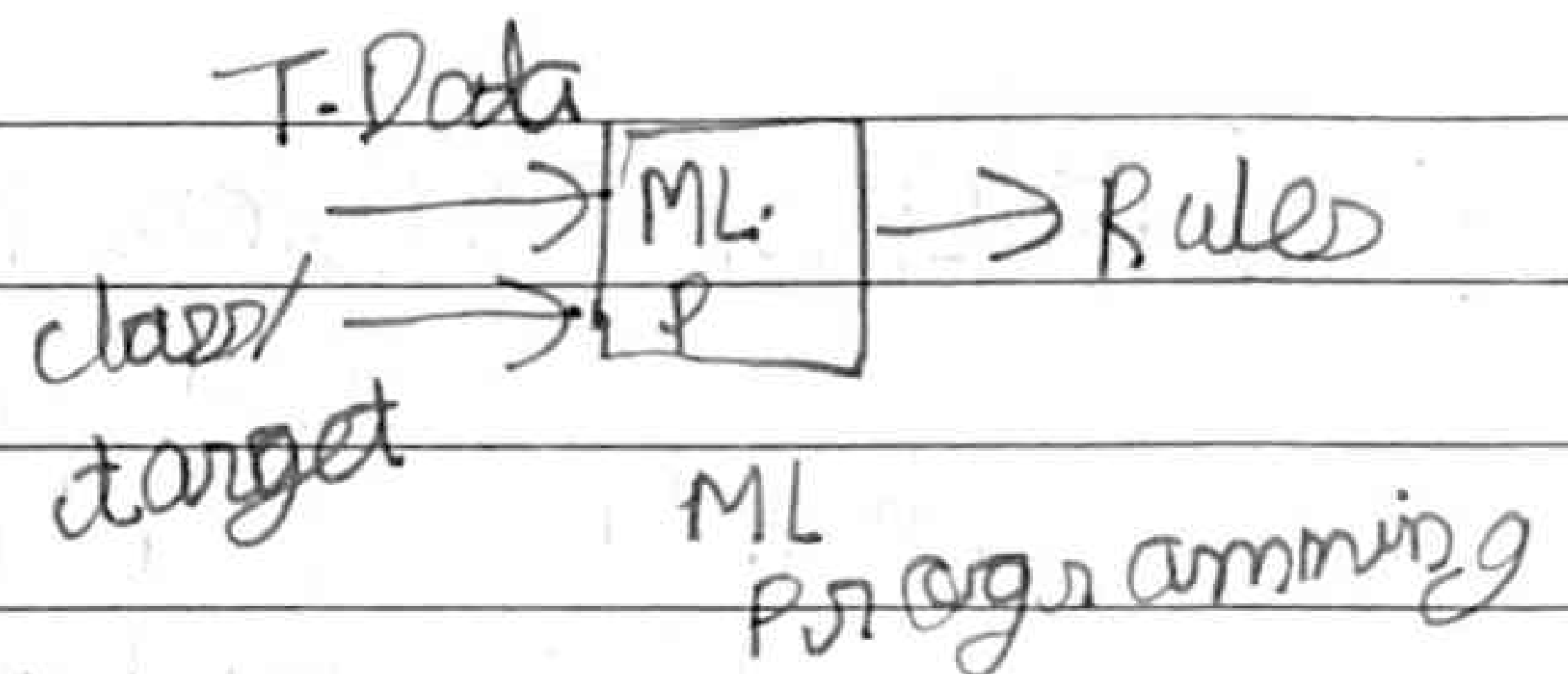
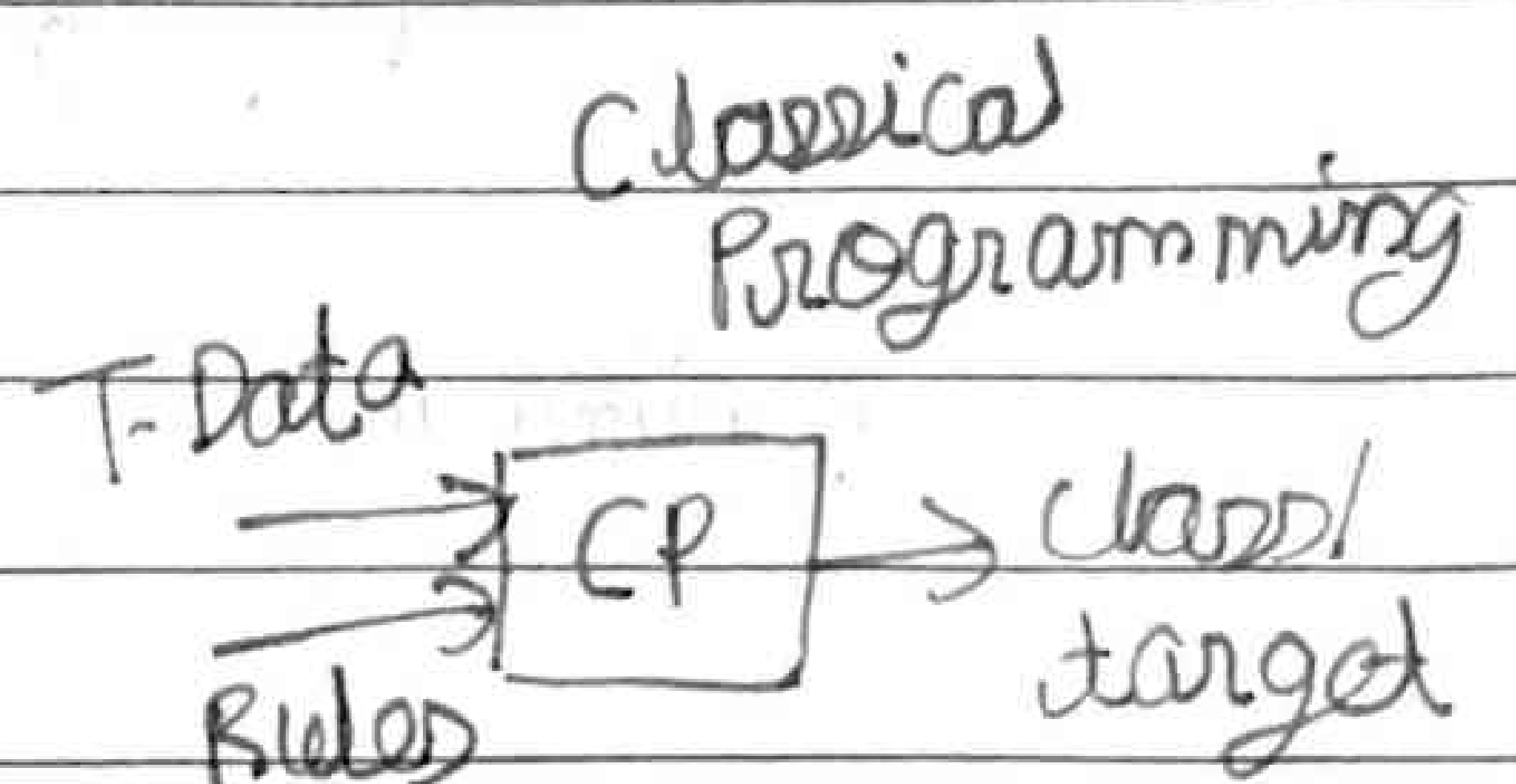
(i) Precision = $\frac{TP}{TP+FP}$ First row

(ii) Recall = $\frac{TP}{TP+FN}$ First column

(iii) F1 = $\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

(iv) Accuracy = $\frac{TP+TN}{TP+FN+TN+FP}$

(v) Specificity = $\frac{TN}{TN+FP}$ 2nd column



Error = 1 - Accuracy

Stages & Tasks in CRISP-DM -

- Determine business objectives
- Collect initial data, describe data, explore data, verify data quality
- Select data, clean data, construct data, integrate data, format data
- Evaluate results
- Plan deployment

Life Cycle of a data mining project -

- 1) Initiation/Business understanding:
Define the project scope, objectives & requirements
- 2) Data Collection and preparation:
Gather and preprocess the data
- 3) Model building:
Develop and train data mining models
- 4) Evaluation:
Test & validate the models
- 5) Deployment:
Implement the models in a production environment
- 6) Monitoring & Maintenance:
Continuously monitor model performance & make necessary updates



Skills needed for Data Mining -

- 1) Statistic & mathematical knowledge
- 2) Programming Skills
- 3) Domain expertise
- 4) Data manipulation & analysis
- 5) Machine Learning
- 6) Critical thinking & Problem solving

Understanding your Data -

1) Data exploration:

- inspect data types (numerical, categorical, text, time series)
- check data distribution (mean, median, mode, range, SD)
- analyze frequency of values in categorical variables

2) Identify Data Quality Issues:

- missing values (detect where & how much data is missing)
- outliers (identify extreme values that may affect model performance)
- duplicates and inconsistencies
(Spot & correct repeated or conflicting entries)

3) Data relationships & Correlations:

- Analyze relationships between variables (eg correlation matrices)
- Use cross tabulations or group statistics for categorical features

4) Data visualizations:

- Used charts & plots to understand trends & patterns:
 - Histograms for distributions
 - Box plots for outliers
 - Scatter plots for relationships
 - Heatmaps for correlations

5) Feature relevance:

- Identify which variables are likely to influence the target variable
- Perform univariate & multivariate analyses to select meaningful features

6) Target Variable Analysis:

- Understand behavior of the variable you want to predict
- For classification: check class balance
- For regression: examine value ranges & trends

Data mining

Predictive

→ Prediction

→ Classification

→ Regression

→ Time Series analysis

Descriptive

→ Association Rules

→ Affinity analysis

→ Sequence analysis

→ Link analysis

→ Clustering

→ Segmentation

→ Outlier analysis

→ Sequence discovery

→ Summarization

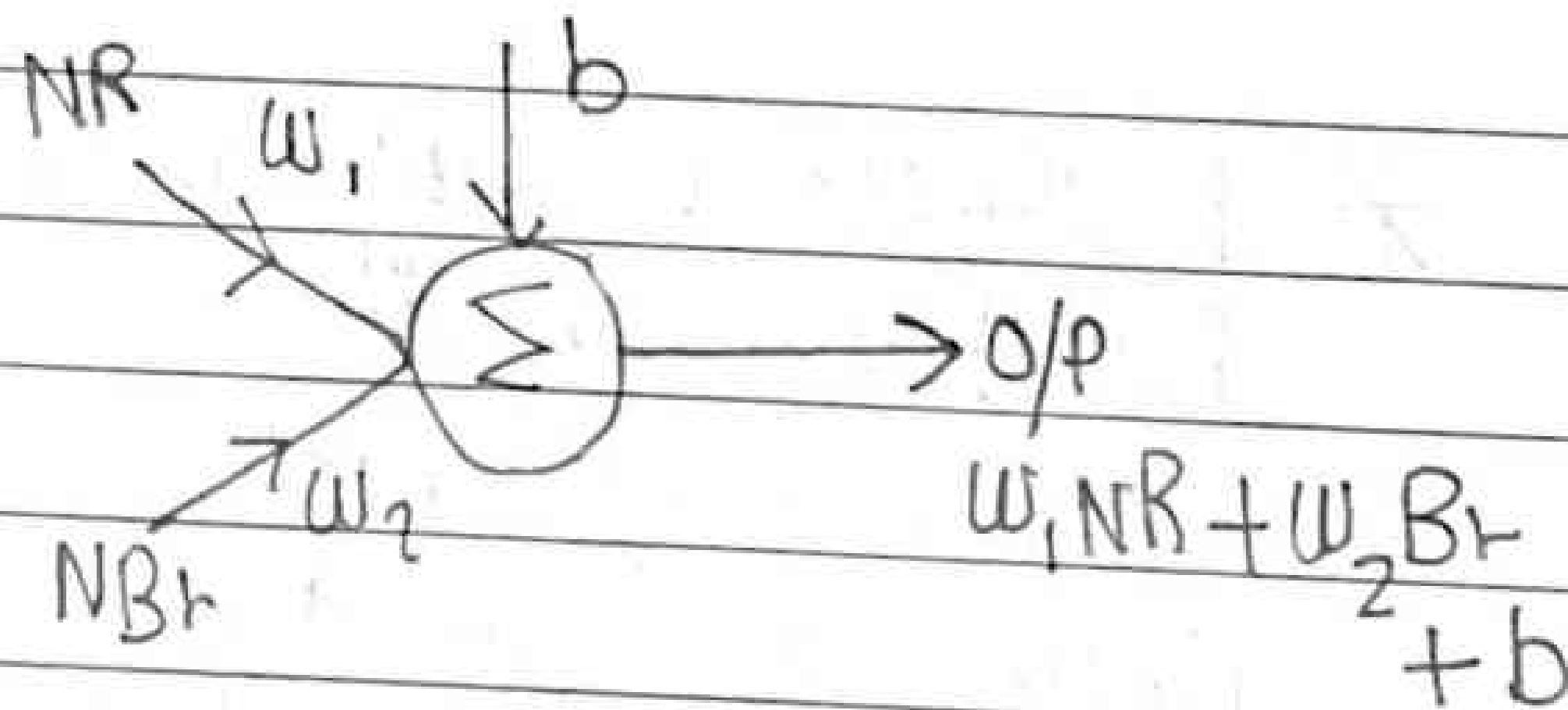
Simplest type of neural network.
Binary classifier

bias is a constant value added to output shift up or down

Single Perceptron

$f(NR, NB)$

	x_1	x_2	
	$w_1 = 0.7$	$w_2 = 0.3$	
$y = mx + c$	NR	NB	Price
	2	2	25 lac
$y = wx + b$	2	3	25.5 lac
	3	2	36 lac

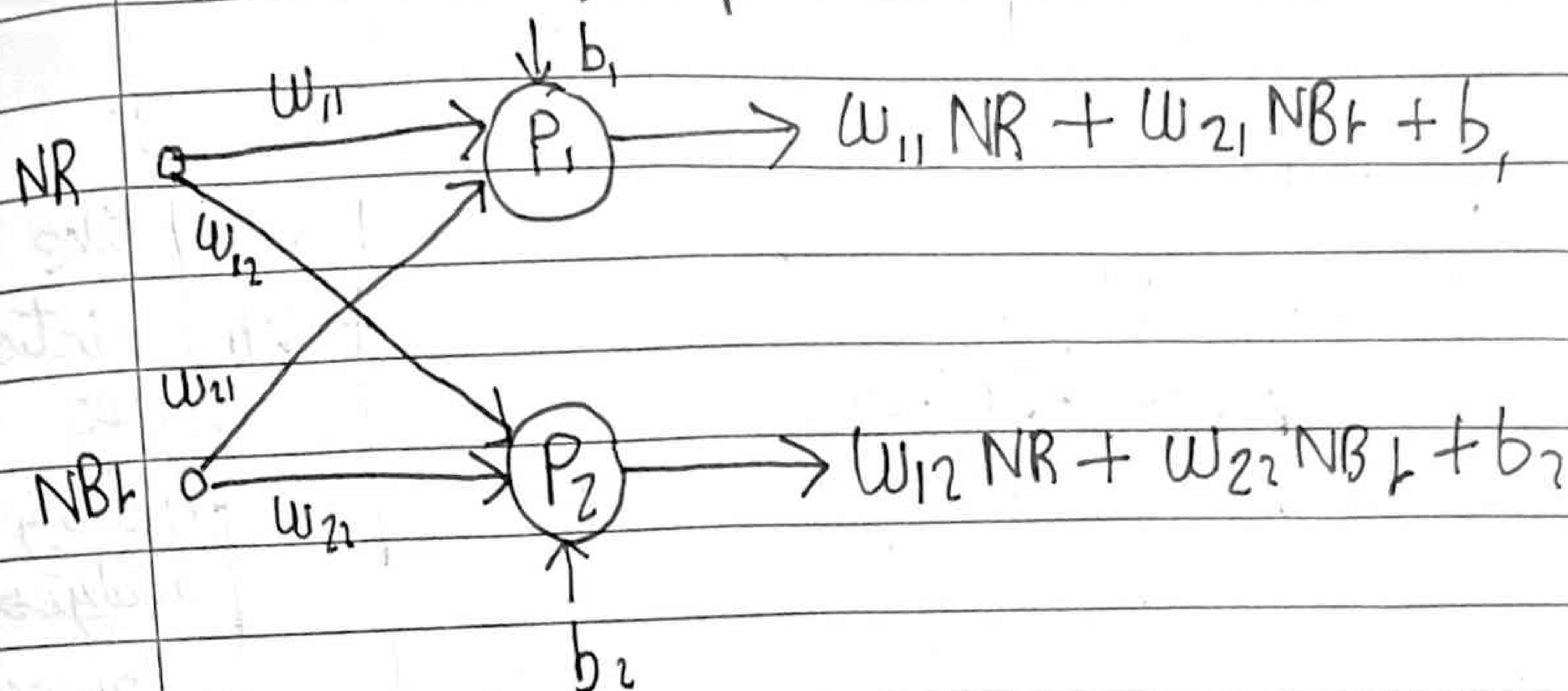


	O/P == 1	NR	NB	Predicted Price
if print (1st)		2	2	$1.4 + 0.6 = 2$
else print (2nd)		2	3	$1.4 + 0.9 = 2.3$
		3	2	$2.1 + 0.6 = 2.7$

weight is a no. that represents how imp a feature is

↑ weight
↑ influence

Multi Neuron perceptron



3 layers:

- Input: Receives the raw data (features). Each neuron in this layer corresponds to one input feature.
- Hidden: Intermediate layers that process the inputs. They apply weights, biases, & activation functions to learn patterns. You can have 1 or more hidden layers.
- Output: Reduces the final result.

Reduces the final result.

For regression, output is a number.

For classification, output is a class label.

* A single perceptron does not have a hidden layer. Perceptron directly connects the input to the output using weights & bias.

It can only model linearly separable problems like classifying data that can be separated with a straight line.

Activation ~~network~~ function: mathematical function applied to the output of a neuron to decide:

- (i) Whether the neuron should be activated
- (ii) How much signal to pass forward

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Graphs & Data visualization-

1) Bar Graph (Bar chart):

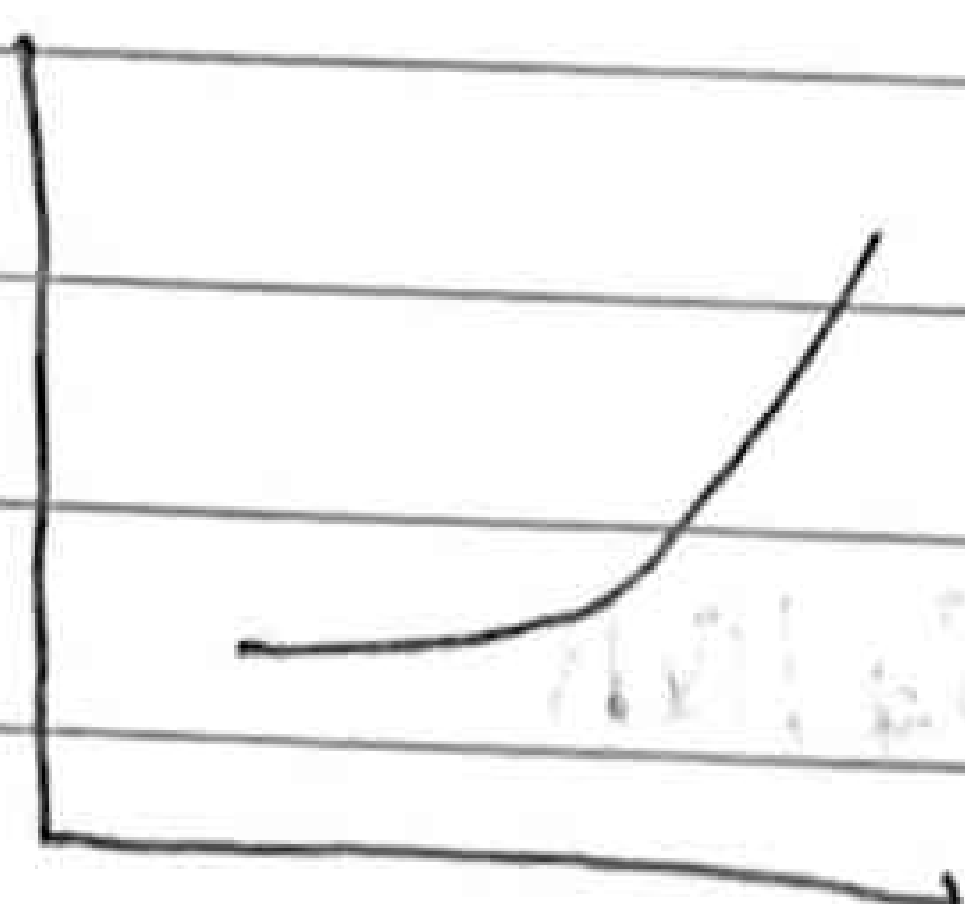
~~Use~~ Compare quantities across distinct categories, sales of different products, population by state



2)

Line Graph:

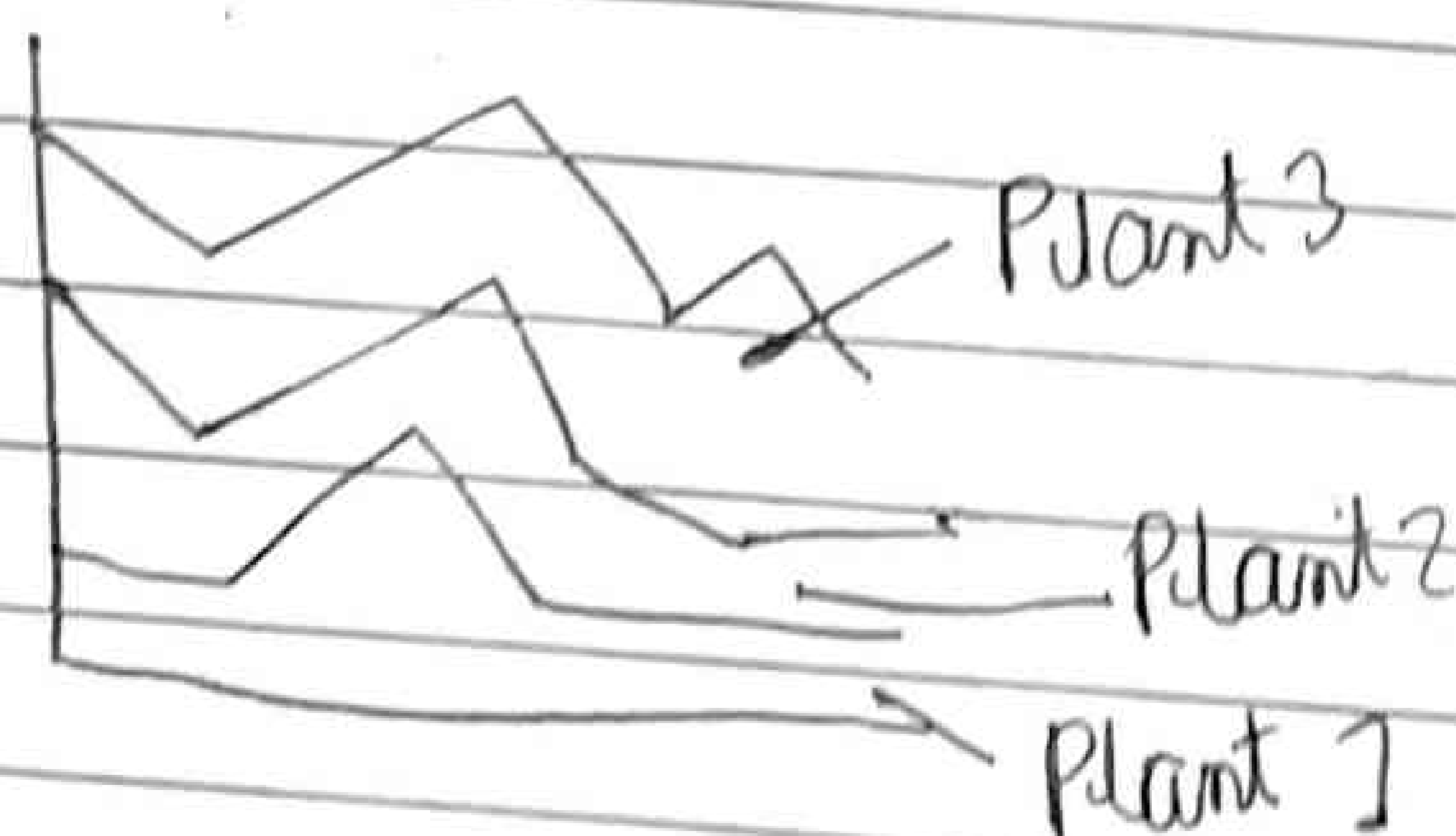
Track trends over time, compare multiple series, detect patterns, fluctuations, and growth



3)

Area Graph:

Show cumulative totals or trends, visualize how different components contribute to the whole over time



4)

Pie Chart:

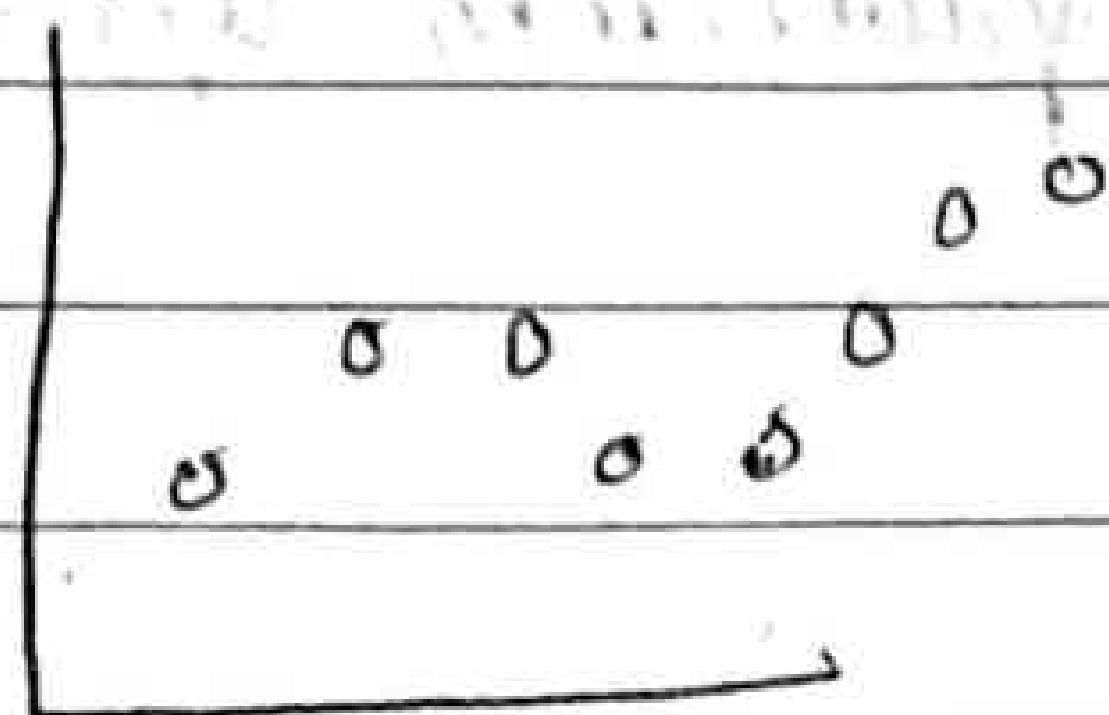
Show part to whole relationships, visualize proportions of different categories.

Limitations: not ideal for comparing small differences, hard to read with many slices



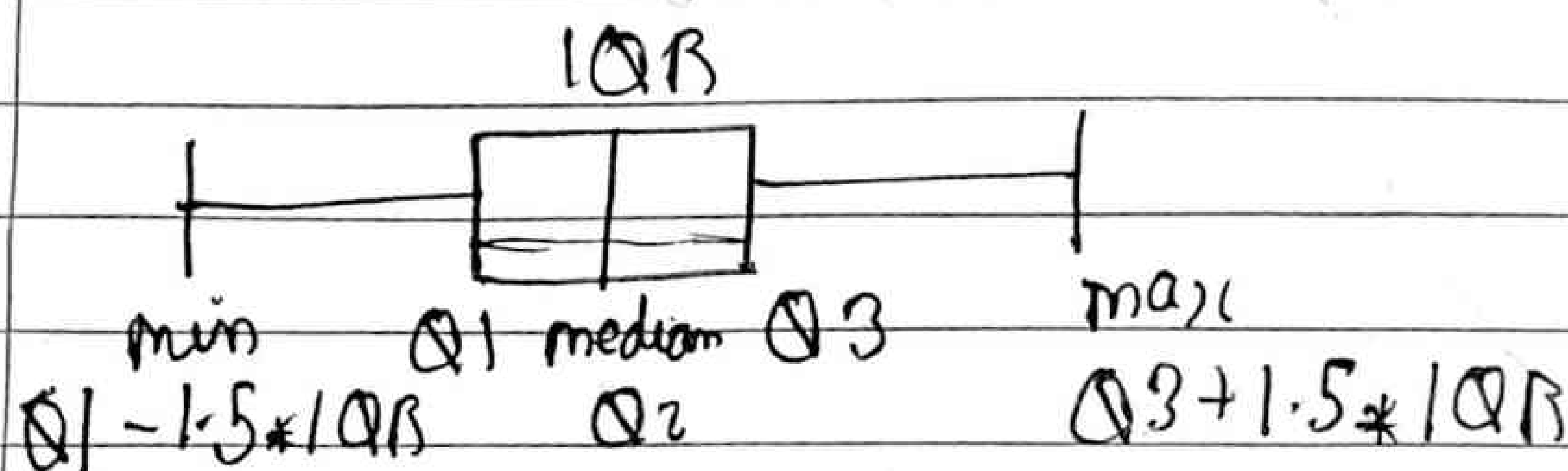
5) Scatter Plot:

Explore correlation or relationship between 2 variables, identify clusters, outliers or trends



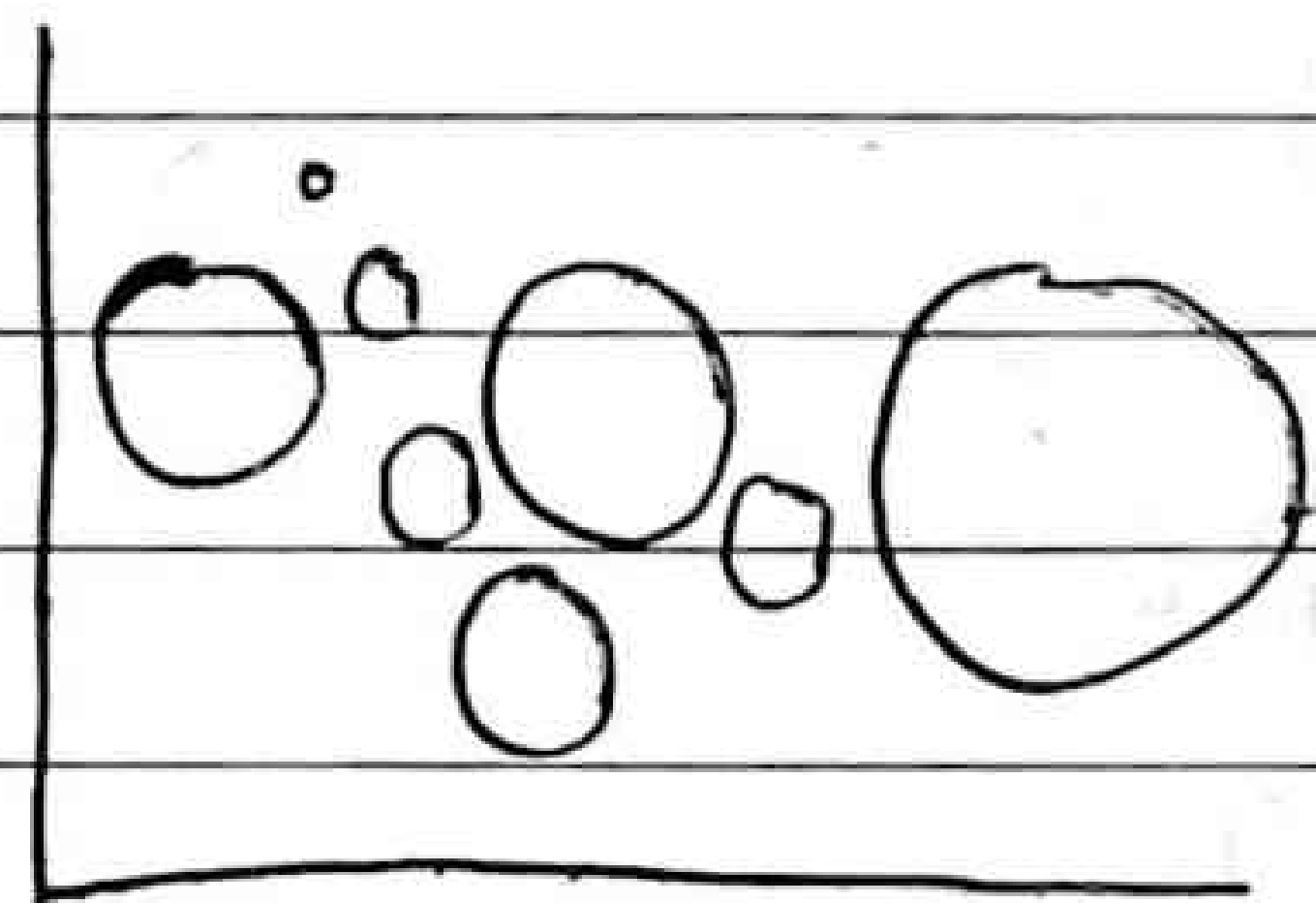
6) Box Plot:

Show distribution, median, quartiles, outliers, and range of data; compare distributions between groups; identify outliers & skewness



7) Bubble chart:

Extension of a scatter plot, each point is a bubble with size representing a third variable; show relationship between 3 variables; visualize impact vs size metrics



8) Heatmap:

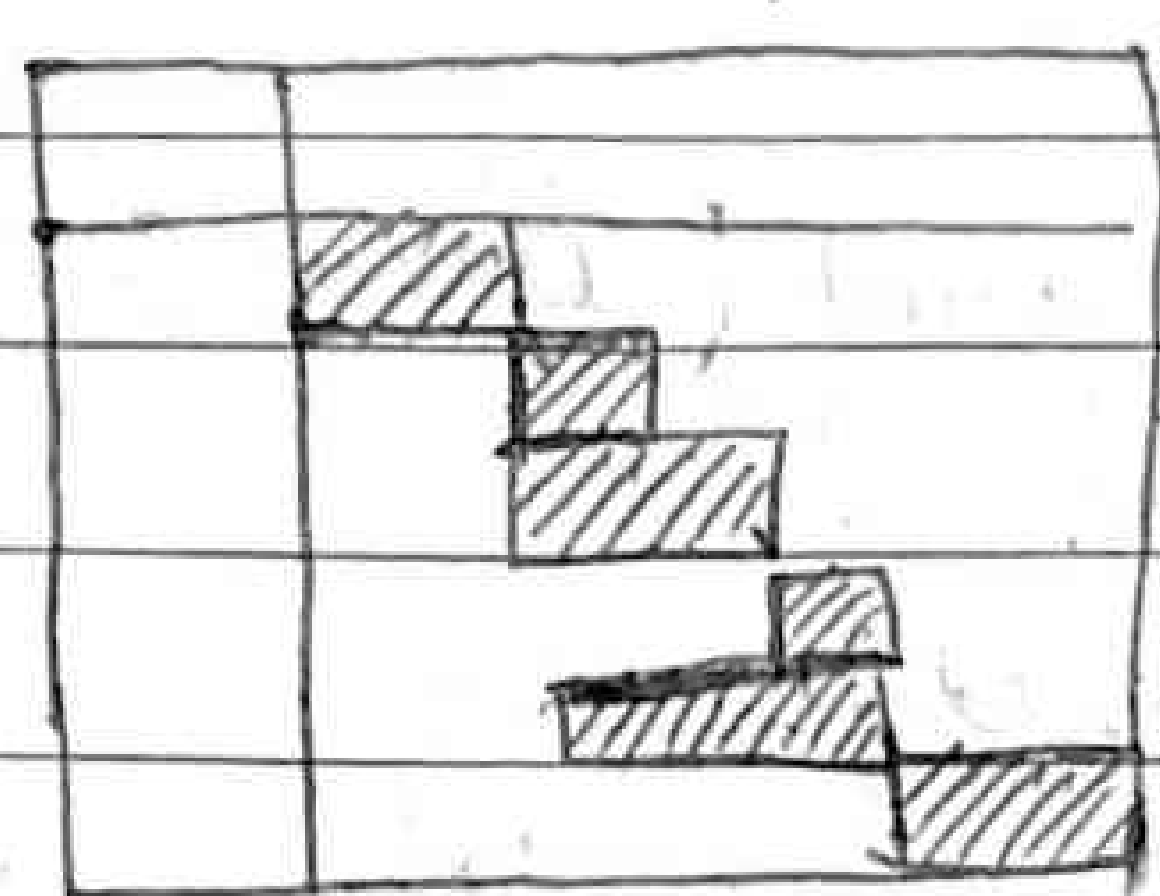
Colour coded matrix representing data values; colour intensity reflects magnitude; analyze patterns, correlations & density



9) Waterfall chart:
Visualize incremental changes that lead to a final result

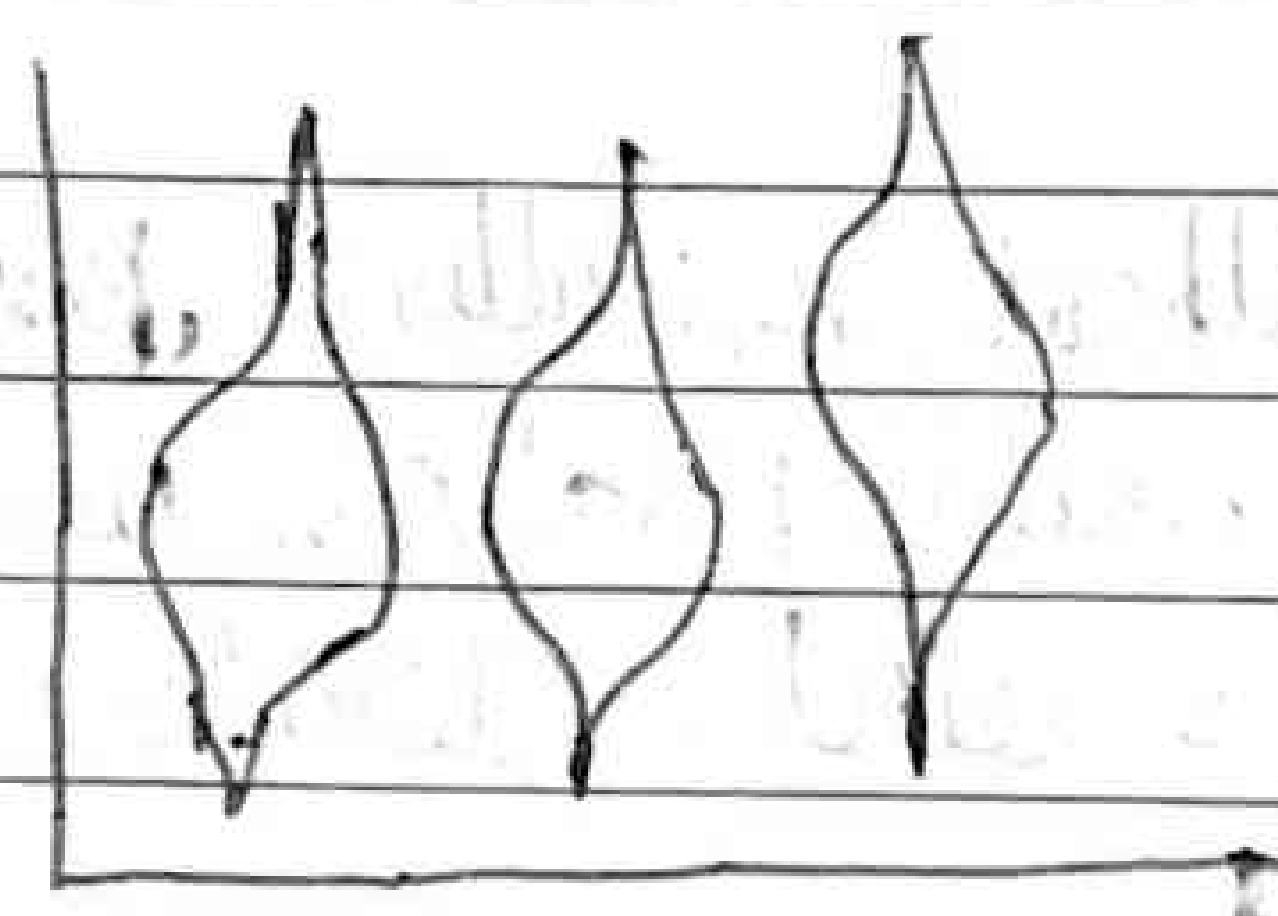
10) Gantt chart:

Horizontal bars represent tasks & timelines; project management to track task durations, dependencies, & schedules



11) Violin Plot:

Combines box plot & density plot; shows distribution shape of data across different groups; better insight into data distribution than boxplot



Definitions:

- 1) Knowledge = Info converted into knowledge about historical patterns & future trends
- 2) Data Warehouses & retrieval: A process of centralized data management
- 3) Architecture of a typical data mining systems: components include a database, data warehouse (DW) or other info repository, DB server, DM engine, pattern evaluation model & user interface

Functionalities of Data Mining

- 1) Characterization & discrimination
- 2) Mining frequent patterns
- 3) Associations & correlations
- 4) Cluster analysis
- 5) Outlier analysis
- 6) Evolution analysis

Characterization -

- Like writing a profile of a sports team
- Produces a summary of the general features or properties of a target class of data objects
- Focuses on characteristics of groups

Discrimination -

- comparing 2 smartphone models
- Compares the general features of the target class with those of one or more contrasting classes
- Checks how does a group differs from others

Cluster analysis:

Analyzing data objects without consulting known class labels to identify homogeneous subpopulations, useful for explanatory data mining & outlier detection

Outlier analysis -

Identifying & examining data objects that significantly differ from the rest of the dataset

Market Basket Analysis -

- Finds frequent itemsets in customer transactions
- Extracts association rules & sequential patterns
- Uses support & confidence to measure rule strength
- Used for marketing, fraud detection, shelf planning, web layout

Types of Missing Data-

1) Missing Completely at Random (MCAR):

Reasons like lab errors, road accident or unintentional skipping of questions; probability of missing data does not depend on income & age

2) Missing at Random (MAR):

Ignorable missingness; probability of missingness on Y does not depend on Y itself after controlling for other variables

3) Missing Not at Random (MNAR):

Nonignorable missingness; probability of missingness on Y depends on values of Y itself

Handling Missing Data-

1) Ignoring missing data

2) Using all available data

3) Imputation

4) Sensitivity Analysis

Single Imputation:

Using a central tendency measure (mean for continuous, median for ordinal, mode for nominal), Regression, Maximum Likelihood, Expectation Maximization (EM) algorithm

Data Structuring & Unit of Analysis

- Data Organization Rules:

Comparison of non rectangular & rectangular data structure

- Unit of observation: objects observed for systematically collected info

~~Frequent P~~

Frequent Patterns = Patterns that occur frequently in data

Association Analysis:

Determining items frequently purchased together
eg. customers who buy computer also buy software
Used in Market Basket Analysis

Evolution Analysis

Analyzing patterns that change over time

Predictive Mining tasks:

Perform inference on current data to make predictions

Classification vs Prediction (Regression)

Aspect	Classification	Prediction
1) Output type	Categorical (labels/classic)	Numerical (continuous values)
2) Example	Spam vs not spam	Predicting house price
3) Goal	Assign data to predefined outer groups	Estimate future/unknown numeric value
4) Algorithms	Decision Tree, Naive Bayes, SVM	Linear Regression, ARIMA

Types of Data Collection

1) Primary data collection

2) Secondary data collection

3) Automated & Technology Driven Collection

↳ uses tools or sensors to gather data automatically

Types of Data (Storage)-

1) Structured data:

- Organized & stored in fixed format
- Pre defined schema, stored in relational databases
- Easy ease of analysis
- Excel, CSV, SQL

2) Semi-Structured data:

- Does not have a rigid schema but still has some structure through tags, markers or meta data
- Self describing structure
- Moderate ease of analysis
- JSON, XML documents

3) Unstructured data:

- Does not follow any predefined format or structure
- No predefined schema or structure
- Harder ease of analysis
- Images, videos, audios, etc

Types of data measurement scales -

Scale	Description	Examples	Key Feature
1) Nominal	Categories, no order	Gender, City, color	Labels only
2) Ordinal	Categories with order	Rank, Grades, Satisfaction	Order matters
3) Interval	Numeric, equal intervals, no true zero	Temperature, IQ, Year	Differences meaningful
4) Ratio	Numeric, equal intervals, true zero	Age, weight, height	Ratios meaningful

* # SPSS

- One of the 3 veterans of data mining market that launched the CRISP-DM in late 1996.
- Industry standard analytics platform

* - Nodes:

Individual components that perform specific operations like data input, transformation, modeling, and output in predictive analytics tools.

* - Streams:

Workflows composed of interconnected nodes that define the sequence of operations from data input to output.

Streams enable users to manage & visualize flow of data through various processing & modeling stages, helping organize complex workflows.

Basically, it provides a visual programming interface to create data streams by dragging & interconnecting nodes. Data flows from one node to the next, depicted by arrows.

SPSS Modeler: User Interface & Workflow

1) Menu Bar:

Top row: File, Edit, Insert, View, Tools, etc

2) Tool bar / Tool icons:

Below menu bar: Open, Save, Run, Stop, Play, Zoom

* 3) Node Palette:

Bottom: tabs like favourites, sources, record ops, field ops, graphs, modelling, output, etc — each contains different nodes to drag into the canvas

* 4) Stream Canvas / Workspace:

Center area where process flow is built with nodes like SQL, Merge, Type, Graphs, etc

5) Output Pane:

Right side: list outputs like tables, histograms, distributions

6) CRISP-DM Project Pane:

Right side, below outputs: shows project stages

7) Status Bar:

Bottom: shows version info, memory usage, status messages

* # Types of Nodes:

Node Type / Category	Specific Node (Examples)	Functionalities
1) Data Input	File, Database, Web Data	Import data from different sources
2) Data Preprocessing	Field Reorder, Missing value handling, Filter, Select fields, Transformation	Data cleaning, transformation, preprocessing
3) Data Audit	Data Audit Node	Descriptive stats, detect missing values, outliers, invalid data, suggest preparation actions
4) Data Imputation	File Node	Custom handling / replacement of missing values
5) Modeling	Logistic Regression, Decision Tree, K-means	Build predictive or descriptive models
6) Type	Type Node	Defines roles of variables: target, input, both or none

7) Output (Report & Stats)	Statistics Node, Table, Chart Export	Calculate stats summary, frequency distributions, generate reports & visualizations correlation
8) Scoring	Apply model, score nodes	Use trained model to make predictions on new data
9) Graphs	Graphs Nodes	Visualizations
10) Evaluation	Evaluation Node	Assess model performance, compare accuracy
11) Workflow management	Super Node	Bundle nodes into reusable components, simplify complex streams
12) Derived from Output	Generated Nodes	Create new nodes from results for deeper analysis
13) Utility	Merge, Append, Partition	Manage data streams
14) Export	Database export, excel, csv	Save processed data or results to external storage

Steps to Explore -

- 1) Start with an empty stream
- 2) Add a data source node (eg- File, database)
- 3) Apply data cleaning nodes (eg- Field reclass, data audit, missing value handling)
- 4) Use Modeling nodes (eg- Logistic regression, Decision Trees, K means clustering)
- 5) Add Output nodes (eg- table, chart, export)

Creating Streams:

Keep clean, modular, documented, efficient, avoid redundancy & errors

Placing Nodes:

Drag from palette to workspace; organize logically

Managing Nodes:

Rename, delete, edit properties, use annotations

Managing Connections:

Link outputs to inputs; define data flow

Generating Nodes from Output:

Create new nodes from existing outputs

Running Streams:

Execute the stream, results appear in output nodes or can be saved

Setting Unit of Analysis:

Define research question, entity, and scope

Creating Datasets:

Collect & reshape data to fit the unit of analysis

Distincting Records:

Ensure uniqueness by removing duplicates (via primary keys)

Aggregating Records:

Summarize multiple records (sum, average, group, time)

Setting Flag Fields:

Create binary fields (true/false) for categorization or filtering

Data Mining Project Framework:

Steps → import data, prepare data, build model, evaluate results

Modeling Objectives:

Build simplified models to analyze, predict, or explain; balance accuracy, simplicity & interpretability

Types of Models:

~~Modeler~~ Classification (Decision Trees, SVM, Naive Bayes, Logistic Regression), Rule Induction, Statistical (Linear/Logistic Regression, Discriminant), Machine Learning, Segmentation (K means, hierarchical)

Model Nugget:

Stores trained model + settings + outputs for reuse without retraining

Evaluating Classification Models:

Metrics → Accuracy, Precision, Recall, F1 score

Cluster Profiles:

Describe clusters with ~~set~~ summary stats

Data Structures & Unit of Analysis-

- Dataset organization Rules: Data can be rectangular (rows = cases, columns = variables) or non rectangular (hierarchical, relational)
- Rectangular Data Structure: Standard tabular form with rows & columns. Easy to analyze but less flexible for complex data
- Unit of Analysis: Main entity being studied (individual, group, event, time, area)
- Unit of Observation: what is ~~exactly~~ actually recorded. Sometimes they are the same, sometimes different.

Identifying Units of Analysis:

- Social Entities = people, groups, organizations
- Time = Years, months, events
- Space = Geographical areas
- Importance = Ensures clarity of what is being studied
- Substituting Units = Shifting focus
- Complexity = Multiple entities, hierarchical (student in class in school), crossed (student x subject)

Data Structure:

- Flat / Rectangular Matrix: Variables as fields; cases as rows
- Adding Complexity: Hierarchical (nested), relational (linked tables), compound / crossed (multiple entities in one record)

Required Unit of Analysis:

- Entity you want results for
- Types: individual, group, event, spatial, temporal
- Steps: Define research question → decide scope → match dataset structure

Methods to Create Datasets (Unit of Analysis):

- 1) Data collection
- 2) Transformation (convert structure to rectangular)
- 3) Flattening: collapse multiple observations into one
- 4) Handling multiple observations: aggregate or partition is needed

Record Handling:

- 1) Distinct Records
- 2) Aggregating Records: Summarize
- 3) Setting Flag Fields: Create binary fields for conditions, filters, or categories

Field Details:

- Field Storages:
 - + Static: Fixed values
 - + Dynamic: Derived/computed
 - + Numeric/Text/Binary: Data type storage
- Field Measurement Labels (Done before)
- Storage ~~vs~~ Measurement Level:
 - + Storage = how data is kept (type)
 - + Measurement = what data means (scale)
- Fields instantiation: defining ~~or~~ fields, inputting values, ensuring data integrity